

Global Air Quality Analysis

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Introduction

- Motivation: Air pollution is a major global health risk
- Goal: Explore trends & make insights accessible via dashboard

Project Overview

- Dataset: WHO Ambient Air Quality Database (2024 update) ~77k records
- Focus pollutants: PM2.5, PM10, NO₂
- Objectives: clean & analyze, identify trends, build interactive dashboard
- Real-world problem: Policymakers and public need clear insights

Solution & Benefits

- Python for cleaning, EDA, statistics, clustering
- Dashboard in Tableau with KPIs, maps, and trends
- Benefits: interactive, filterable, highlights improving vs stagnating regions
- Clear tool for policymakers, NGOs, researchers, and public

Potential Users

- Policymakers & governments
- NGOs & health organizations
- Researchers & students
- General public

Demonstration

- Dashboard features:
 - KPI cards (Avg PM2.5, PM10, NO₂)
 - Global map: city-level air quality
 - Line chart: pollutant trends over time
 - Regional comparison bar chart
- Interactive filters: Year, Region, Pollutant

Limitations

- Annual averages only (no daily/hourly data)
- Missing data in some countries/years
- No direct health outcome data (hospital admissions, mortality)

Future Steps

- Link air quality data with health datasets
- Add socioeconomic factors (GDP, urbanization)
- Forecast pollution trends with time-series ML
- Deploy dashboard publicly for NGOs & policymakers

Thank you!